Yoav Gudes

AI/ML Engineering Co-op Student
University of Waterloo 2028
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Summary

- Nanotechnology Engineering student with 16 months of co-op experience across AI/ML, data engineering, and GPU testing.
- Experience developing and optimizing ML models, including NLP classifiers and gradient boosting algorithms.
- Proficient in Python, SQL, and LLM integration, with experience building automated pipelines and agentic AI workflows.

Education

Nanotechnology Engineering: Quantum Engineering Option | **University of Waterloo** **Sept 2023 – Apr 2028**

- **Relevant Coursework:** Calculus I–IV, Linear Algebra, Differential Equations, Numerical Methods, Probability & Statistics

Work Experience

Process Engineering Intern | WOLF Advanced Technology **Sept 2025 – Apr 2026**

- Assembled and diagnosed 50+ NVIDIA/AMD GPUs, validating bandwidth and thermal stability with CUDA and FurMark.
- Led stack-up failure analysis across 10 failed GPU units, tracing root causes to assembly and design issues.

AI/ML Engineering & Data Science Intern | TD Bank **Jan 2025 – Apr 2025**

- Developed an NLP classifier for client-call sentiment analysis, benchmarking an in-house solution against Azure AI.
- Built an Azure Databricks pipeline (speech-to-text, utterance parsing) and hand-labeled a 10,000+ utterance dataset.
- Encoded utterances with MPNet sentence-transformer embeddings; trained XGBoost classifiers with class rebalancing.
- Optimized XGBoost using Bayesian optimization (Optuna), improving F1 score from 0.71 to 0.90.
- Presented the proof-of-concept model to senior staff, showing a custom model was a viable alternative to Azure AI.

Data Engineering Intern | MedReddie **May 2024 – Aug 2024**

- Built an AI-powered scraping pipeline using LLM APIs, BeautifulSoup, and Selenium to automate supplier data collection.
- Collected contact details for 8,000+ suppliers and structured $50M+ in product data, replacing manual data entry.
- Generated 1,000+ due-diligence questionnaires (DDQs) with MedReddie's CoPilot AI for hospital purchasers.
- Uploaded product, DDQ and supplier data to staging and production environments using SQL and Python.

Technical Skills

Machine Learning

- *ML Algorithms:* XGBoost, Random Forests
- *NLP & Embeddings:* MPNet Sentence Transformers, UMAP
- *Model Optimization:* Bayesian Optimization, GridSearchCV

Programming & Platforms

- *Languages:* Python, SQL, Java, MATLAB
- *Libraries:* Pandas, scikit-learn, XGBoost, Optuna
- *Platforms:* Azure Databricks, Jupyter, Git, GitHub

LLMs & Generative AI

- *API Integration:* API-Based LLM Calls in Python Pipelines
- *Document Generation:* Automated Reports & Summaries
- *Agentic Workflows:* Multi-Step Task Automation

Data Engineering

- *Scraping:* BeautifulSoup, Selenium, Requests
- *Data Quality:* Fuzzy Matching, Deduplication, Validation
- *Databases:* SQL, Staging & Production Environments

Leadership & Awards

President | Hillel Waterloo & Laurier **May 2025 – Aug 2026**

- Led student programming and oversaw executive board operations across a multi-campus organization.

Awards

- **University of Waterloo:** Term Distinction (Dean's List), Excellent Academic Standing & Co-op Review, Jordan Banks Award.
- **TD Bank:** Achieve the Dream Canada Scholarship, Co-op Challenge Finalist